BP DUAL-CHOKEPOINT

Energy Shock Equity Stress Model

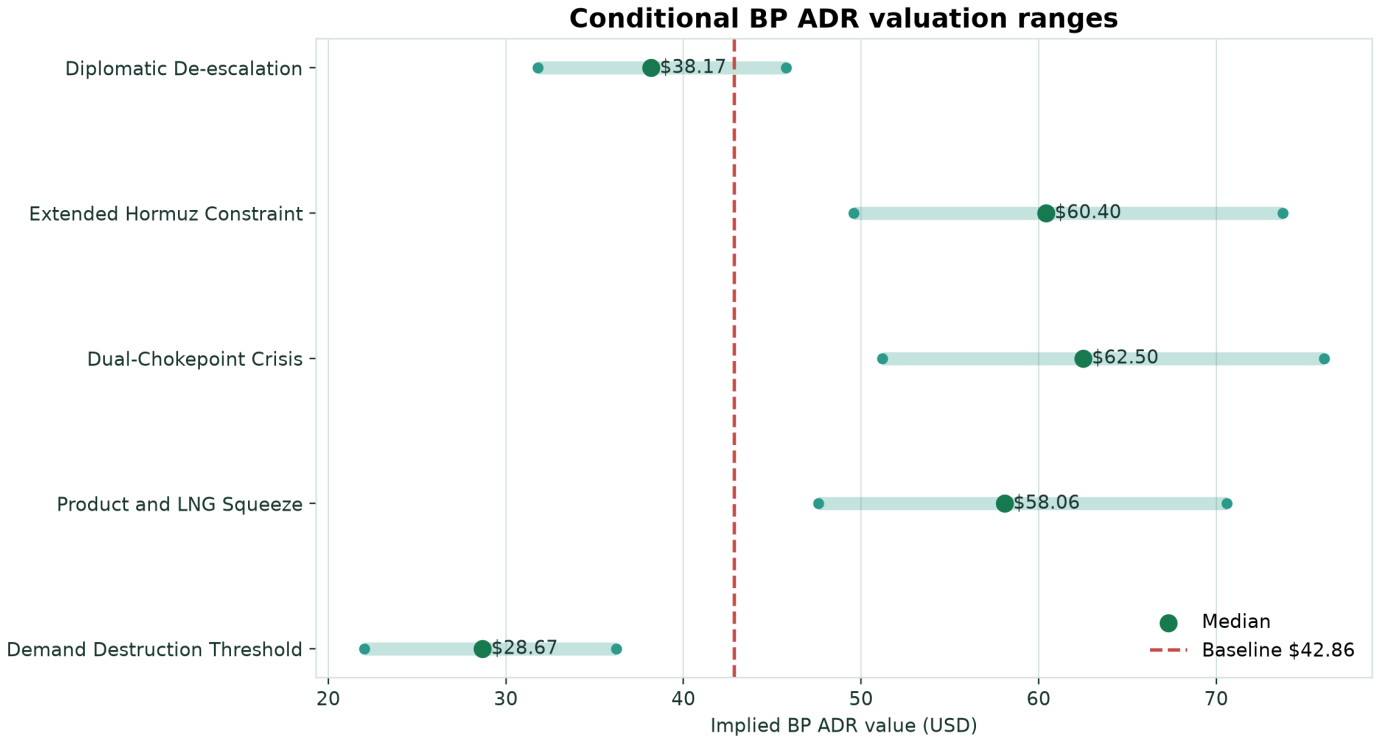
A scenario-based assessment of simultaneous Strait of Hormuz and Bab al-Mandeb disruption, refined-product pressure, LNG scarcity, and demand destruction.

Market baseline: 2026-07-27 | BP ADR $42.86 | Brent $90.93/bbl

Executive judgment

BP benefits initially from higher oil and gas realizations, tighter refining margins, and trading optionality. The benefit is nonlinear and conditional. A long dual-chokepoint disruption increases freight, security and working-capital costs while raising the probability of volume loss, demand destruction, balance-sheet pressure, and valuation-multiple compression.

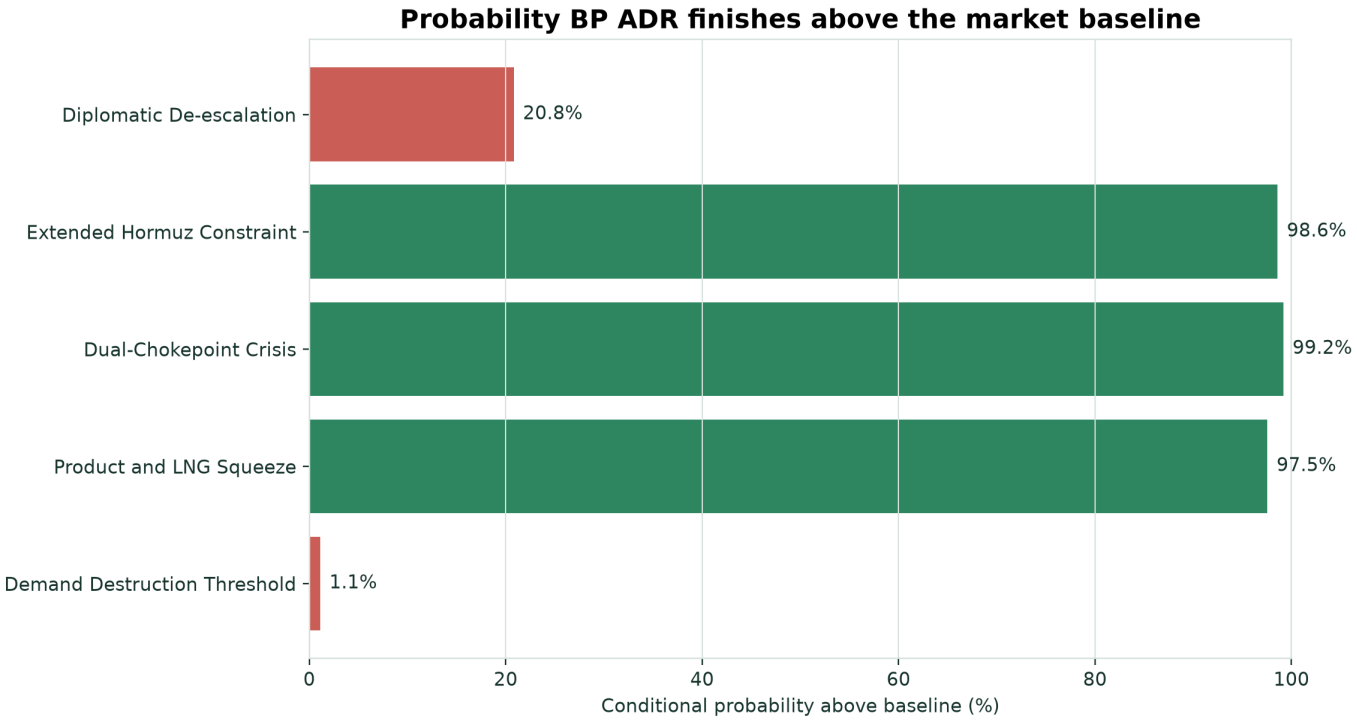
Highest median scenario	Demand-threshold median	Dual-crisis probability
$62.50	**$28.67**	**99.2%**
Dual-Chokepoint Crisis	P(>base) 1.1%	above baseline



Scenario results

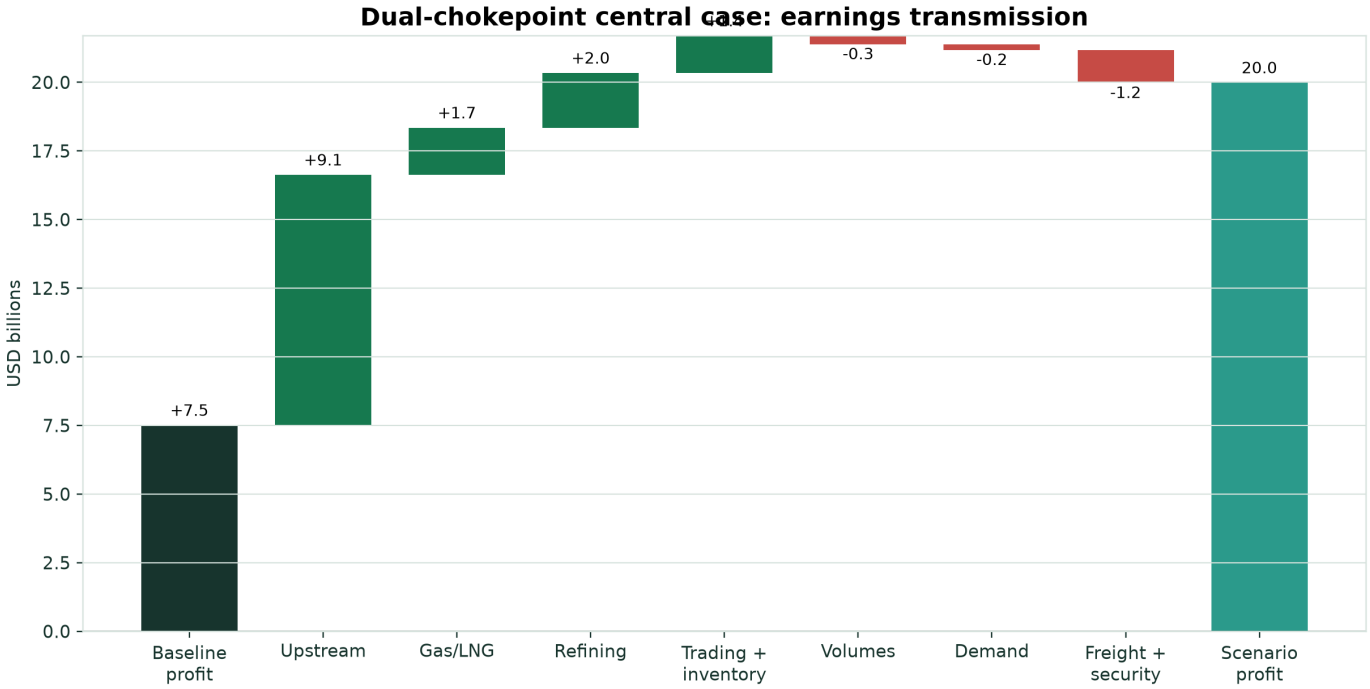
Each distribution is conditional on the specified scenario. It does not estimate the probability that the geopolitical scenario itself will occur.

Scenario	Brent	TTF	Net income	P10	Median	P90	Above base
Diplomatic De-escalation	$82	$14.4	$6.0B	$31.80	$38.17	$45.78	20.8%
Extended Hormuz Constraint	$126	$26.1	$16.6B	$49.60	$60.40	$73.74	98.6%
Dual-Chokepoint Crisis	$139	$30.3	$19.9B	$51.21	$62.50	$76.07	99.2%
Product and LNG Squeeze	$117	$23.7	$15.0B	$47.60	$58.06	$70.60	97.5%
Demand Destruction Threshold	$81	$21.9	$5.7B	$22.01	$28.67	$36.22	1.1%



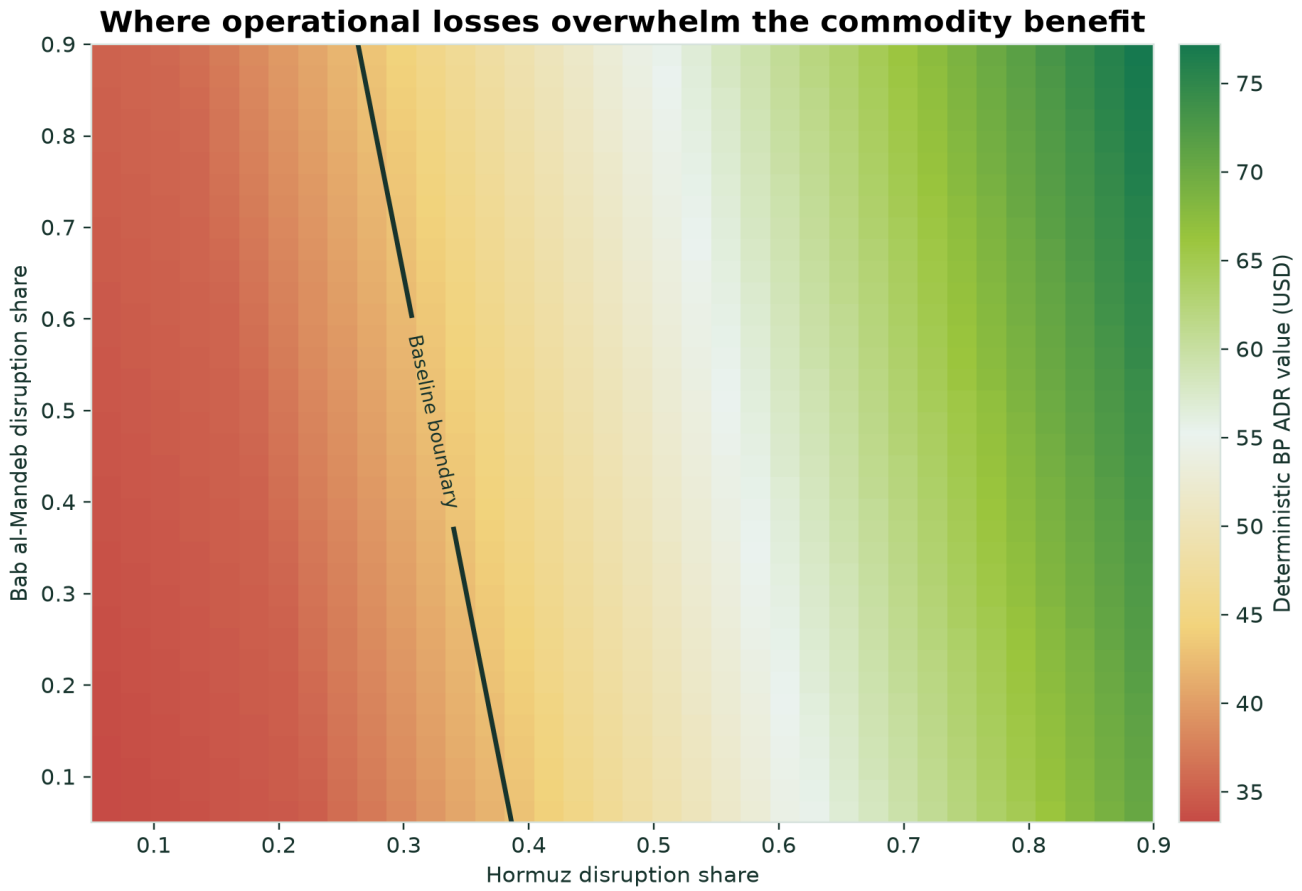
Why BP's response becomes nonlinear

Positive channels	Negative channels
Higher crude realizations	Cargo delays and potential volume losses
Higher international gas and LNG prices	Freight, insurance and security costs
Tighter refining margins	Working-capital requirements and net debt
Trading optionality and inventory gains	Customer demand destruction
Geographic upstream diversification	Recession and equity-risk-premium expansion



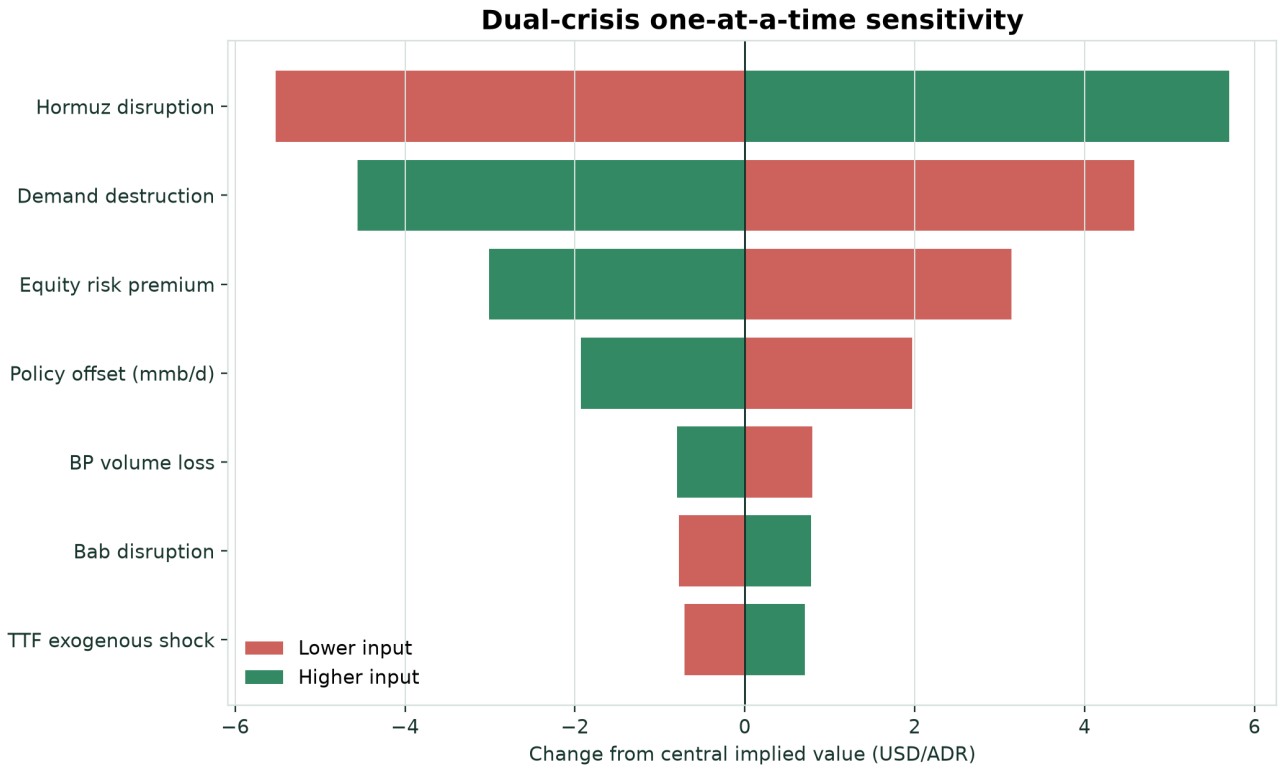
The waterfall shows the transparent central assumptions for the dual-chokepoint case. Monte Carlo results add uncertainty around physical losses, capture rates, trading, working capital, market returns, valuation weights, and risk-premium compression.

Threshold map



The dark boundary marks the modeled combination at which BP's deterministic ADR value crosses the current-share-price baseline. All other assumptions are held at the dual-crisis reference case, including 9% demand destruction and a 12% equity risk premium.

Sensitivity and interpretation



Hormuz severity is the largest one-at-a-time driver because it changes both physical crude availability and BP's realized-price benefit. Demand destruction and the equity risk premium work in the opposite direction. Bab al-Mandeb has a smaller direct physical effect because rerouting is possible, but it remains important through freight, insurance, delays, and refining logistics.

Method and limitations

Physical supply layer. Gross Hormuz oil disruption is combined with effective Bab al-Mandeb losses and Russian product losses, then reduced by available bypass capacity, policy releases, and demand response.

Market layer. A convex short-run supply-gap function estimates Brent. LNG interruption affects a TTF reference price, while product losses, logistics pressure, and demand destruction affect a composite refining crack spread.

Corporate layer. Commodity prices flow through BP's upstream, gas/LNG, refining, trading, customer, volume, freight, security, and working-capital channels. Trading is bounded because public reporting cannot reproduce BP's internal positions.

Valuation layer. The implied ADR value is an ensemble of earnings-scaling, free-cash-flow, and market-factor estimates, with net-debt, GBP/USD, and risk-premium effects. Each scenario uses 10,000 Monte Carlo draws.

Important limitations

This is a transparent research stress test, not BP's internal planning model, an event forecast, or investment advice. Scenario probabilities are not assigned. Public segment disclosures cannot identify trading books, cargo-level exposure, hedges, tax effects, or asset outages in real time. Results should be recalibrated as filings and event data change.

Primary sources

- BP - 2025 results, 2025 Annual Report and Form 20-F, and Q1 2026 results.
- U.S. EIA - Hormuz chokepoint, Red Sea shipping, and LNG disruption analyses.
- Reuters - July 27, 2026 oil-market and Hormuz traffic update.
- FRED and yfinance - optional live market observations when available.

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